Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT | Paper -2(Adv-2019-P2-Model) | Date: 07-12-2025

Time: 02.00Pm to 05.00Pm | GTA-07 | Max. Marks: 186

KEY SHEET

PHYSICS

1	**AD**	2	**ABCD**	3	**ABCD**	4	**AC**	5	**ABCD**	6	**AD**
7	**ABD**	8	**AC**	9	**8**	10	**4.20**	11	**70**	12	**14**
13	**1.20**	14	**5**	15	**D**	16	**A**	17	**A**	18	**C**

CHEMISTRY

19	**CD**	20	**ABCD**	21	**AC**	22	**BC**	23	**AC**	24	**ABD**
25	**ABD**	26	**AB**	27	**15.00**	28	**4.00**	29	**295.50 - 295.60**	30	**58.38 - 58.40**
31	**2.30 - 2.40**	32	**95.30 - 95.40**	33	**C**	34	**C**	35	**A**	36	**C**

MATHEMATICS

37	**BCD**	38	**ABC**	39	**ABC**	40	**AC**	41	**ABCD**	42	**ABCD**
43	**ABCD**	44	**BC**	45	**5**	46	**2**	47	**4**	48	**4**
49	**0**	50	**1**	51	**C**	52	**D**	53	**B**	54	**C**

For More Join: @IITJEE_Advanced

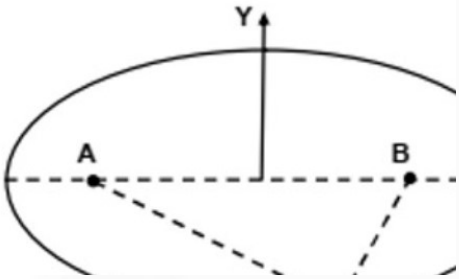
SOLUTIONS

PHYSICS

01. As $PA+PB=10\,\ell=$ Constant

The bead will move along the elliptical path in vertical plane.

Such that $2a=10\,\ell$ and $2ae=8\ell \Rightarrow a=5\ell, b=3\ell$



$\therefore$ Equation of ellipse is

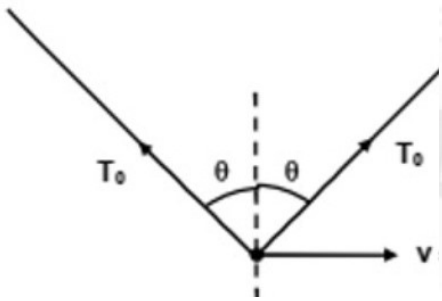
$$\frac{x^2}{(5\ell)^2}+\frac{y^2}{(3\ell)^2}=1$$

$$\frac{r_0=\left[1+(y')^2\right]^{3/2}}{|y''|} \text{ at } (0,-3\ell) \qquad \left(\because y'=0, y''=\frac{3}{25\,\ell}\right)$$

$$\therefore r_0=\frac{25\ell}{3}$$

$$2T_0\cos\theta-mg-\frac{m}{r_0}\left(\sqrt{2g\ell}\right)^2 \qquad \left(\because \cos\theta=\frac{3}{5}\right)$$

$$T_0=\frac{31\,mg}{30}$$



02. $40\times 4\cos 30^0=(40+20)u_c$

$$\frac{160}{60}\times\frac{\sqrt{3}}{2}=u_c$$

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$$u_c = \frac{4}{\sqrt{3}}$$

$$\frac{1}{2}\times 40\times\left(\frac{4\sqrt{3}}{2}\right)^2 = \frac{1}{2}\times 60\times\left(\frac{4}{\sqrt{3}}\right)^2 + \frac{1}{2}\times 6000x^2$$

$$40\times 4\times 3 = \frac{60\times 16}{3} + 6000x^2$$

$$480\left(1-\frac{2}{3}\right) = 6000x^2$$

$$x = \sqrt{\frac{160}{6000}} = \sqrt{\frac{2}{75}}$$

After coming back,

$$40\times\frac{4}{\sqrt{3}} = 40v_p + 20v_c$$

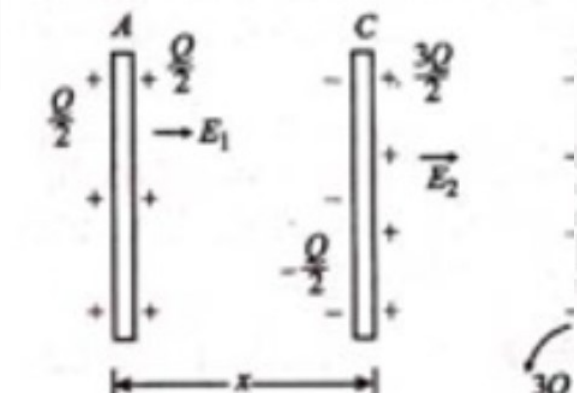
$$e = 1 = \frac{v_c - v_p}{4/\sqrt{3}}$$

$$\frac{8}{\sqrt{3}} = 2v_p + v_c$$

$$-\frac{4}{\sqrt{3}} = -v_c + v_p$$

$$v_p = \frac{4}{3\sqrt{3}}; v_c = \frac{16}{3\sqrt{3}}$$

03. Charge distribution on the three plates will be as shown.



This will ensure that the electric field is zero inside the metal plates.

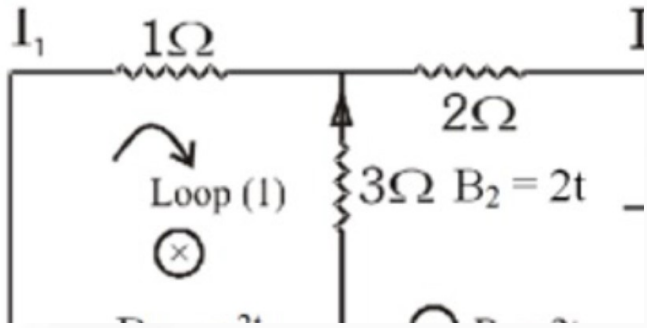
Potential difference between A and B is

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$$P_{AB} = E_1 x + E_2 x + \frac{Q}{2A\varepsilon_0}x + \frac{3Q}{2A\varepsilon_0}x = \frac{2Q}{A\varepsilon_0}x$$

Before insertion of plate C, the potential difference was same.

04. For loop $1\ \phi_1 = 0.2 \times e^{-2t}$



$$E_1 = -\frac{d\phi_1}{dt}\Big|_{t=0} = 0.2(-2)e^{-2t}\Big|_{t=0} = -0.4\,vot$$

$\Rightarrow 0.4V$ in Anti clockwise

$$E_2 = -\frac{d}{dt}(0.2 \times 2t) = 0.4$$ in clockwise direction

Now in loop 1

$$0.4 - I_1 - 3(I_1 + I_2) = 0$$

$$\Rightarrow 0.4 = 4I_1 + 3I_2 \qquad \ldots..(1)$$

in loop 2

$$0.4 - 0.3 = 3(I_1 + I_2) + 2I_2$$

$$0.1 = 3I_1 + 5I_2 \qquad \ldots..(2)$$

by $3 \times (1) - 4 \times (2)$

$$1.2 = 12I_1 + 9I_2$$

$$0.8 = 0 = 11I_2$$

$$\Rightarrow I_2 = \frac{0.8}{11} = \frac{8}{110}A$$

$$\text{now } I_1 = \frac{0.1 - 5I_2}{3} = \frac{\frac{1}{10} - 5\left(-\frac{8}{110}\right)}{3}$$

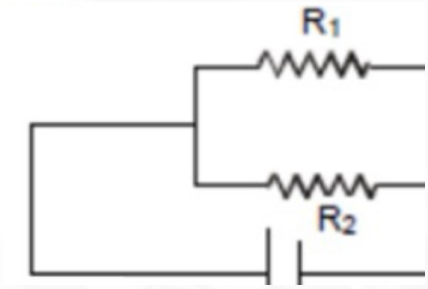
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$$I_1 = \frac{\frac{1}{10} + \frac{40}{110}}{3}$$

05. $\frac{(200)^2}{R_1} \times t_1 = 2 \times 6000 \times 20 \Rightarrow \frac{(200)^2}{R_1} = 400\ watt \Rightarrow R_1 = 100\,\Omega$

$\frac{(400)^2}{R_2} \times t_2 = 4 \times 12000 \times 20 \Rightarrow \frac{(400)^2}{R_2} = 800\ watt \Rightarrow R_2 = 200\,\Omega$

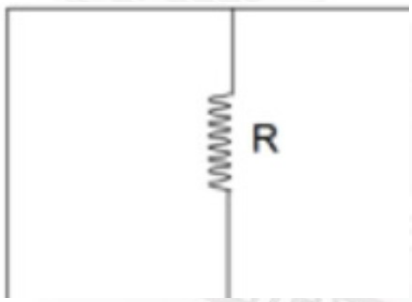
In series $\frac{(300)^2}{R_1 + R_2} \times t = 2 \times 6000 \times 30 \Rightarrow t = 20$ min



In parallel $\frac{(300)^2}{R_1R_2 / R_1 + R_2} \times t = 4 \times 12000 \times 27$

$t = 16$ min

06. Calculate the time constant



Battery is replaced by a wire $\tau = \frac{L}{R_{ab}} = \frac{L}{R}$

07. Process AB is isothermal so $T_A = T_B = T_0$

Now $\Delta U_{BC} = \frac{nfR(T_C - T_B)}{2}$

{we need to calculate T_C}

Process CA is adiabatic

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$\Rightarrow T_C.V_C^{\gamma-1} = T_A.V_A^{\gamma-1}$

$\Rightarrow T_C.(2V_0)^{5/3-1} = T_0(V_0)^{5/3-1}$

$\Rightarrow T_C = \frac{T_0}{2^{2/3}} = \frac{2T_0}{3}$

Now $\Delta U_{BC} = \frac{1\times 3\times R\left(\frac{2T_0}{3} - T_0\right)}{2} = \frac{-RT_0}{2}$

Option A is correct

$\rightarrow$ Work done for all processes

$AB \rightarrow W_{AB} = nRT_0 \ln\left(\frac{2V_0}{V_0}\right)$

$= 1\times RT_0 \ln 2 = 0.7RT_0$

$\Delta U_{AB} = 0$ & $Q_{AB} = \Delta_{AB} + W_{AB} = 0.7RT_0$

$BC \rightarrow W_{BC} = 0, \Delta U_{BC} = -RT_0/2$

$Q_{BC} = W_{BC} + \Delta U_{BC} = -RT_0/2$

$CA \rightarrow W_{CA} = -\Delta U_{CA}$

$= -\frac{1\times 3\times R\left(T_0 - \frac{2T_0}{3}\right)}{2} = -\frac{RT_0}{2}$

No heat is supplied ion CA

Now net work done

$\Rightarrow W = W_{AB} + W_{BC} + W_{CA}$

$= 0.7RT_0 + 0 - 0.5RT_0$

$\Rightarrow W = 0.2RT_0 = RT_0/5$

Heat is supplied to gas only in process AB that is $0.7RT_0$

Now efficiency $= \frac{\text{Work Done}}{\text{Heat Supplied}}$

$= \frac{2RT_0}{7RT_0} = \frac{2}{7}$

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08. $\left|\frac{dN}{dt}\right| = \lambda N$

$A = \log\left|\frac{dN}{dt}\right|$

$A_0 = \log \lambda N$

$A = \log \lambda N$

$A = \log \lambda + \log N$

$A = \log N + \log \lambda$

$Y = mx + c$

$N = N_0 d^{-\lambda t}$

$\log N = \log N_0 + \log e^{-\lambda t}$

$\log N = \log N_0 - \lambda t$

$A = \log N_0 - \lambda t + \log \lambda$

$= -\lambda t + \log \lambda + \log N_0$

Slope $\rightarrow \frac{-A_0/2}{t_0}$

$-\lambda = \frac{-A_0}{2t_0}$

Condt $\longrightarrow A_0$

$\log \lambda + \log N_0 = A_0$

$\log \lambda N_0 = A_0$

$\lambda N_0 = e^{A_0}$

$N_0 = \frac{e^{A_0}}{\lambda}$

$N_0 = \frac{2t_0 e^{A_0}}{A_0}$

09. $\frac{1}{2}mv_0^2 - \frac{GmM}{5R} = \frac{1}{2}mv^2 - \frac{GmR}{R}$ .....(i)

(Conservation of mechanical energy)

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$$\frac{mv_0}{2}5R = mvR \qquad \text{.....(ii)}$$

(Conservation of angular momentum about C)

from (i) and (ii)

$$v_0 = \sqrt{\frac{32GM}{105R}} \Rightarrow k = 8$$

10. For minimum de-Broglie wavelength of electron, the momentum and hence kinetic energy of electron must be maximum

$$13.6\left(\frac{1}{1^2} - \frac{1}{2^2}\right) - \phi = \frac{h^2}{2m\lambda_1^2}$$

$$13.6z^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right) - \phi = \frac{h^2}{2m\left(\frac{\lambda_1^2}{6.1}\right)}$$

$$\frac{13.6 \times \frac{3}{4} - \phi}{13.6z^2 \times \frac{3}{4} - \phi} = \frac{1}{6.1} \qquad \text{.....(i)}$$

For the unknown gas

Energy of first lyman line-energy Balmer series limit $= 2 \times 13.6\, eV$

$$\Rightarrow 13.6z^2\left(\frac{3}{4}\right) - 13.6z^2\left(\frac{1}{4}\right) = 2 \times 13.6$$

$$z = 2$$

So from equation (i)

$$\phi = 4.20\, eV$$

11. Equation b/w A and B; $(50)^2 - (10)^2 = 2as$

Equation b/w B and A; $V_A^2 - (50)^2 = 2as$

12. $$u = -\left(4 + \frac{4}{2}\right) + 2 = -12\, cm$$

$$\frac{1}{v} + \frac{1}{-12} = \frac{1}{-8}$$

$$\Rightarrow v = -24\, cm$$

for final image (refraction through plane surface) we can write

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$$24 = 2\left(\frac{x}{1} + \frac{4}{2}\right)$$

$$\Rightarrow 24 = 2x + 4$$

$$x = 10\,cm$$

so distance from P is 14 cm.

13. $I = I_0 + 9t_0 + 2\sqrt{t_0 9I_0}\cos\phi$

$$I = 10I_0 + 6I_0\cos\phi$$

$$I_{max} = 16I_0, I_{min} = 4I_0$$

$$\frac{16I_0}{4} = 10I_0 + 6I_0\cos\phi$$

$$\cos\phi = -1$$

$$\phi = \pi, 3\pi, 5\pi.....$$

$$\frac{2\pi y}{\beta} = \pi, 3\pi, 5\pi.....$$

So, $y_2 - y_1 = \beta = \frac{D\lambda}{d} = 1.2$ mm

$$a = 1.2\ \text{mm}$$

14. Let equation of the wave be

$$y = A\sin kx\cos\omega t$$

$$\because 5\sqrt{3} = A\sin(k\times 10) \qquad .....(1)$$

and $5\sqrt{3} = A\sin\left(\frac{\pi}{30}\times 10\right)$ using equation (1)

$$\Rightarrow A = 10\,mm \qquad \because k = \frac{2\pi}{\lambda} = \frac{\pi}{30}$$

$$\Rightarrow \lambda = 60\,cm \qquad \because L = n\frac{\lambda}{2} \qquad \Rightarrow n = \frac{2L}{\lambda} = \frac{2\times 150}{60} = 5$$

As the number of loops is 5, the string vibrates in the 4th overtone 5th harmonic.

15. Nuclear radioactivity and nuclear reactions theory

16. Nuclear radioactivity and nuclear reactions theory

17. Capillarity and angle of contact

18. Capillarity and angle of contact

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CHEMISTRY

19. $V=\dfrac{nRT}{P}$

$V_A = 200R,\ V_C = 400R, V_B = 400R$

$P = \dfrac{400R}{600R - V}$

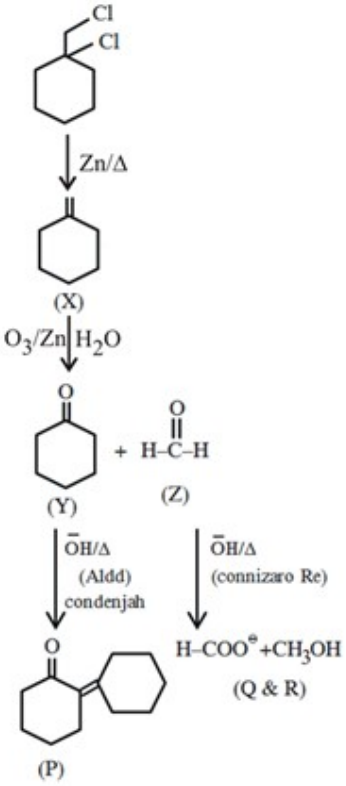
$W_{AB} = -\int Pdv$

$= -400R \ln 2$

$\Delta S = nC_V \ln \dfrac{T_2}{T_1} + nR \ln \dfrac{v_2}{v_1} = nC_p \ln \dfrac{T_2}{T_1} + nR \ln \dfrac{P_1}{P_2}$

$= 0.2L.atm$

20.



21. Net cell reaction is

$\underset{(aq)}{Cd}(s) + 2AgCl(s) \rightarrow 2Ag(s) + Cd^{2+}(aq) + 2Cl^-$

$\Delta G^o_{50°C} = -nFE° = -2 \times 96500 \times 0.6 = -115800J$

$\Delta G^o_{0°C} = -nFE° = -2 \times 96500 \times 0.7 = -135100J$

$\Delta S° = nF\left(\dfrac{E_2^o - E_1^o}{T_2 - T_1}\right) 2 \times 96500 \times \dfrac{0.6 - 0.7}{50}$

$= -386J / K$

$\Delta H° = \Delta G° + T.\Delta S°$

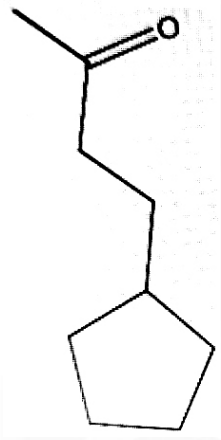
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$= (-135100) + 273 \times (-386) = 240478\text{J}$

22. (D) can give iodoform test

23. $4\pi r^2\Psi^2$ vs r plotted for a H-orbital curve contains 3 maxima and 2 angular node then orbital will be 5d. There will be minimum four atoms of hydrogen atom require during de-excitation from 4th shell to 1st shell to cover all distinct spectral line.

24. Of the two anionic $LiAlH_4$ and $NaBH_4$ complexes, AlH_4^- has much the higher hydride donor ability. Both copper pyrite and copper glance are sulphide minerals of copper

25.  C give above compound

26. Compounds (I) and (III) will reduce Fehling's solution and will also mutarotate Compounds (IV) will reduce Tollen's reagent but not mutarotate

27. (i) $Ag_2SO_3 + H^+ \rightarrow 2Ag^+ + SO_2 + H_2O$

(ii) $FeSO_4$ solution absorbs NO and becomes brown because of the formation of $\left[Fe(NO)\right]^{2+} SO_4^{2-}$

(iii) Being covalent do not dissociate

(iv) Ammonium hydroxide in presence of ammonium chloride is group reagent of III group

(v) $2\left[Cr(OH)_4\right]^-$ (green colouration) $+3H_2O_2 + 2OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$

(vi) Red colour solution or precipitate is produced when reagent reacts in alkaline solution.

(vii) $HO.SO_2NH_2 + HNO_2 \rightarrow N_2 + 2H^+ + SO_4^{2-} + H_2O$

(viii) $Fe^{2+} + \left[Fe(CN)_6\right]^{3-} \rightarrow Fe^{3+} + \left[Fe(CN)_4\right]^{4-}$

$4Fe^{3+} + \left[Fe(CN)_4\right]^{4-} \rightarrow 4Fe\left[Fe(CN)_6\right]_3 \downarrow$(Turnbull's blue)

(ix) $4Cr(OH)_3 \downarrow + 5O_2^{2-} \rightarrow 4CrO_4^{2-}$ (yellow solution) $+6H_2O$

$a = 5\ b = 4\ \ \dfrac{a^2 + b^2}{3} = \dfrac{25 + 16}{3} = 13.67$

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28. $\rho_{ideal} = \dfrac{Z \times \dfrac{M_0}{N_A}}{a^3}$

$$\frac{4 \times \frac{62.5}{N_A}}{\left(500 \times 10^{-10}\right)^3} = \frac{4 \times 62.5 \times 1.67 \times 10^{-24}}{125 \times 10^{-24}} = 3.34 g/ml$$

$\rho_{actual} = 3.09 g/ml$

$= 3.34 - 3.09 = 0.25 g/ml$

$= 250 g/l$

$\dfrac{250}{62.5} = 4$

29. $1000 = K_H X_{HCl} \Rightarrow K_H = \dfrac{1000}{0.02} = \dfrac{100000}{2} 750$

$= \dfrac{100000}{2} X_{HCl} \Rightarrow \dfrac{15}{1000} = \dfrac{15}{1000}$

$P_{Hexane} = 300 \times \dfrac{985}{100} = \dfrac{2955}{10} = 295.5$

30. $2Ag^+ 2e^- \rightarrow 2Ag(s)$ $\quad E^0 = 0.80V$

$C_6H_{12}O_6 + H_2O \rightarrow C_6H_{12}O_7 + 2H^+ + 2e^-$ $\quad E^0 = -0.05V$

The net reaction is:

$C_6H_{12}O_6 + H_2O \rightarrow 2Ag(s) + C_6H_{12}O_7 + 2H^+$ $\quad E^0 = 0.75V$

$E_{cell} = E^0 - \dfrac{RT}{nF} \ln K = 0$

or $\ln K = \dfrac{nFE^0}{RT} = 2 \times 0.75 \times 38.92 = 58.38$

31. For $FeSO_4$ $\quad Fe^{2+} \xrightarrow{\text{Oxidation}} Fe^{3+} \xrightarrow{\text{Reduction}} Fe^{2+}$

For FeC_2O_4 $\begin{cases} Fe^{2+} \xrightarrow{\text{Oxidation}} Fe^{3+} \xrightarrow{\text{Reduction}} Fe^{2+} \\ C_2O_4^{2-} \xrightarrow{\text{Oxidation}} CO_2 \uparrow \end{cases}$

Let mM or Meq. Of $FeSO_4$ and FeC_2O_4 be a and b respectively

$\therefore$ Meq. of Fe^{2+} in $FeSO_4$ + Meq. of Fe^{2+} in FeC_2O_4 + Meq. of $C_2O_4^{2-}$ in FeC_2O_4 = Meq.of $KMnO_4$ used

$$a \times 1 + b \times 1 + b \times 2 = 40 \times \frac{1}{15} = \frac{8}{3}$$

$\therefore a + 3b = \dfrac{8}{3}$

32. $\left(Cu^+\right)_2 \rightarrow Cu^{2+} + Cu$

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The solution after dissolution of Cu_2O in dil H_2SO_4 contains Cu^{2+} and Cu ions. Cu^{2+} ions reacts with KI to give CuI_2 which is converted CuI and I_2^-

$$Cu^{2+} + 2I \rightarrow CuI_2 \rightarrow CuI + \frac{1}{2}I_2$$

Millimole of KI taken $= \frac{8.3}{166} \times 1000 = 50$

Now, KI left unused reacts with oxidizing agent to liberate I_2 again.

$$2I^- \xrightarrow{\text{Oxidant}} I_2 \xrightarrow{2Na_2S_2O_3} Na_2S_4O_3 + 2NaI$$

$\therefore$ Millimole of KI left = Millimole of $Na_2S_2O_3$ used

(mole ratio of I^- to $Na_2S_2O_3$ is 1 : 1)

$= 10 \times 1.0 = 10$

Therefore, millimole of KI used for $Cu^{2+} = 50 - 10 = 40$

$\therefore$ Millimole of $Cu_2O = 20$

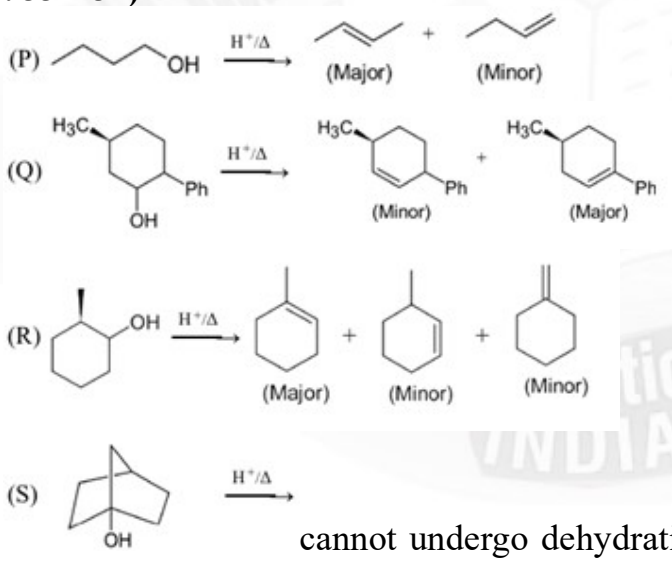
$$\left[\begin{array}{l} Cu_2^+ \rightarrow 2Cu^{2+} \\ \therefore \text{Mole ratio } Cu_2^+ : Cu^{2+} :: 1:2 \end{array}\right]$$

or $\frac{w}{M} \times 1000 = 20$

$\frac{w \times 1000}{143} = 20 \qquad \therefore w_{Cu_2O} = 2.86$

$\therefore \%$ of $Cu_2O = \frac{2.86 \times 100}{3} = 95.33$

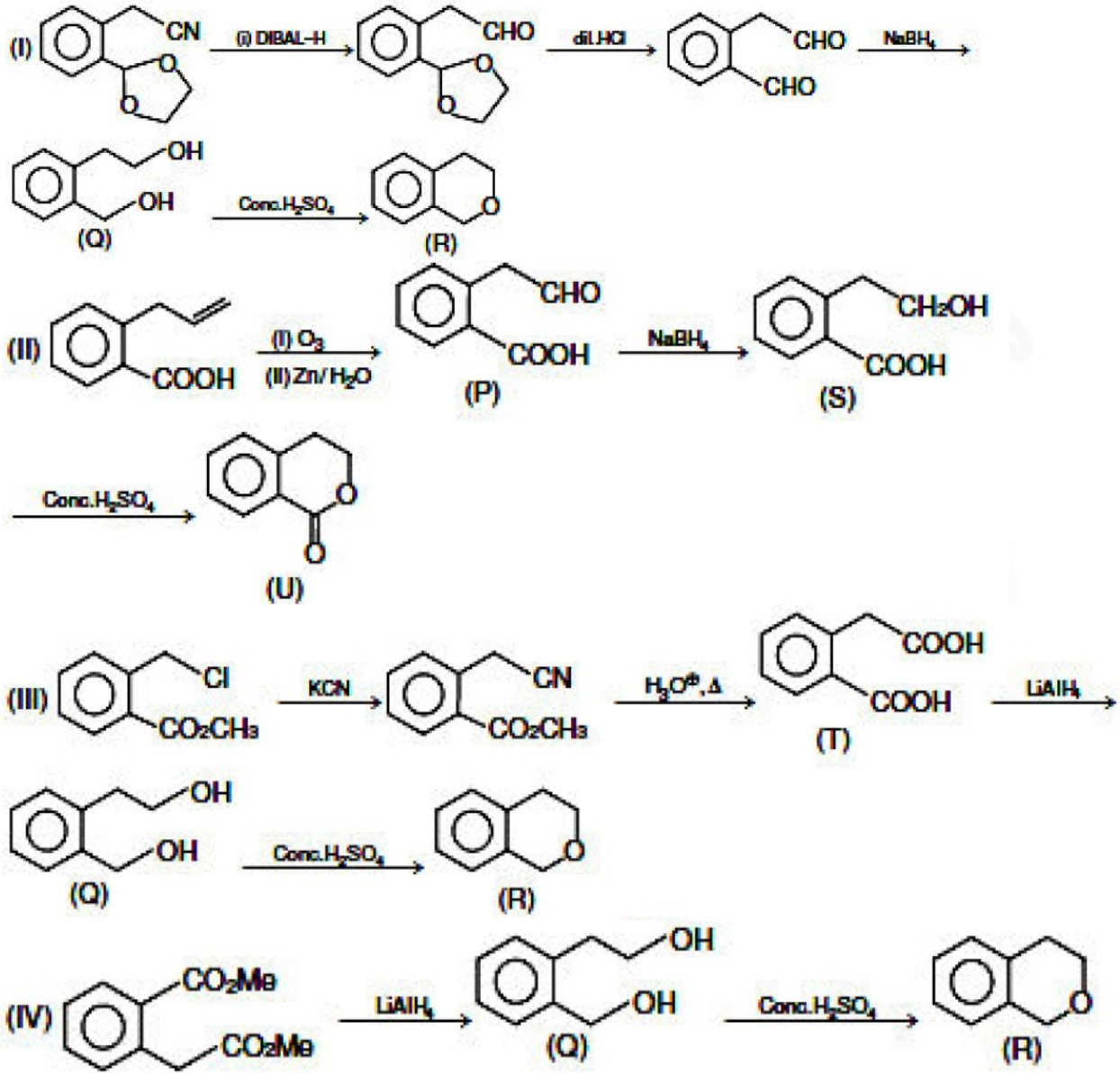
(Q.No. 33 – 34)



(S) $\xrightarrow{H^+/\Delta}$ cannot undergo dehydration as bridge head C-atom can never be sp^2 hybridised

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(Q.No. 35 – 36)



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MATHEMATICS

37. For ellipse $\frac{x^2}{16}+\frac{y^2}{7}=1$

And hyperbola $\frac{x^2}{(144/25)}-\frac{y^2}{(81/25)}=1$

$16-7=16e_1^2$

And $\frac{144}{25}+\frac{81}{25}=\frac{144}{25}e_2^2$

$\frac{3}{4}=e_1$

$e_2=\frac{15}{12}=\frac{5}{4}$

$e_2=\frac{5}{4}$

Foci $=(\pm 3, 0)$

For point of intersection of hyperbola and ellipse

$\frac{x^2}{144}+\frac{y^2}{81}=\frac{1}{25}$

$\Rightarrow y^2\left(\frac{1}{63}+\frac{1}{81}\right)=\frac{1}{9}-\frac{1}{25}=\frac{16}{9\times 25}$

$\Rightarrow y=\pm\frac{3\sqrt{7}}{5}$

$x^2=\left(1-\frac{63}{25\times 7}\right)16=\frac{16\times 16}{25}$

$x=\pm\frac{16}{5}$

Point of intersection $\equiv\left(\pm\frac{16}{5}, \pm\frac{3\sqrt{7}}{5}\right)$

Equation of asymptote $y=\pm\frac{3}{4}x$

Point $P\equiv\left(\pm\frac{16}{5}, \pm\frac{12}{5}\right)$

Sum of square of coordinates of P $=\frac{256+144}{25}=16$

$x^2+y^2=16=a^2$

For More Join: @IITJEE_Advanced

$\Rightarrow$ P lies on auxiliary circle

38. Key : A, B, C

$C_2 = 4C_1; C_3 = 4C_2 = 16C_1$

$R_2 = 6 + R_1;\ R_3 = 6 + R_2 = 12 + R_1$

$R_1 + R_2 + R_3 = 21C_1$

$\Rightarrow 18 + 3R_1 = 21C_1$

$\Rightarrow 6 + R_1 = 7C_1 = R_2$

$$M = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 7 \\ 1 & 4 & 8 \end{bmatrix}$$

39. Since $\sec^2\theta > 1$

$\Rightarrow \left[(n+1)\sec^2\theta\right] > \left[n\sec^2\theta\right]$

Hence, f and g are both one-one

Let $k < n\sec^2\theta < k+1$ and $k < m\,\text{cosec}^2\theta < k+1$

$k\cos^2\theta < n < (k+1)\cos^2\theta$ and $k\sin^2\theta < m < (k+1)\sin^2\theta$

Adding given $k <$ integer $< k+1$

$\Rightarrow A \cap B = \phi$ Suppose $k \notin A$

Suppose $(k+1) < (n+1)\sec^2\theta$

$\Rightarrow n\sec^2\theta < k$ and $(k+1) < (n+1)\sec^2\theta$

$\Rightarrow n < k\cos^2\theta$ and $n+1 > (k+1)\cos^2\theta$

$\Rightarrow k < (k-n)\text{cosec}^2\theta,\ k+1 > (k-n)\text{cosec}^2\theta$

$\Rightarrow \left[(k-n)\text{cosec}^2\theta\right] = k$

$\Rightarrow k \in B$

40. We have $a_r.a_{r+2} = a_{r+1} - 1 \forall r \geq 0$ and $a_0 = 1$

$$a_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta}$$

41. $x\vec{a} + y\vec{b} - \vec{c} = 0$ If $\Rightarrow x + y - 1 = 0$ then C lies on line segment AB

If $x > 0,\ y > 0$ and $x + y - 1 = 0$ then C lies on line segment AB

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If $x<0,\ y>0$ and $x+y-1=0$ then C lies on line segment AB

If $x>0,\ y<0$ and $x+y-1=0$ then C lies on line segment AB

If $x>0,\ y>0$ and $x+y-1<0$ then C lies on line segment AB

42. A) $\int x^2dx+\int d\left(\frac{e^x}{y}\right)=\int 0dx$

$x^3y+3e^x=3cy$

A) put $x=r\cos\theta, y=r\sin\theta \Rightarrow x^2+y^2=r^2; \tan\theta=\frac{y}{x}$

$d\theta=\frac{xdy-ydx}{x^2+y^2}$

$\Rightarrow r^2d\theta=xdy-ydx$

$\therefore \frac{rdr}{r^2d\theta}=\sqrt{\frac{a^2-r^2}{r^2}}$

$\int\frac{dr}{\sqrt{a^2-r^2}}=\int d\theta \Rightarrow \sin^{-1}\left(\frac{x}{a}\right)=\theta+c$

$\sin^{-1}\left(\frac{\sqrt{x^2+y^2}}{a}\right)=\tan^{-1}\frac{y}{x}+c$

C) $\frac{xdy-ydx}{x^2+9y^2}=dx$

$\frac{\frac{xdy-ydx}{x^2}}{1+\left(\frac{3y}{x}\right)^2}$

$\frac{1}{3}\int\frac{d\left(\frac{3y}{x}\right)}{1+d\left(\frac{3y}{x}\right)}=\int dx$

$\frac{1}{3}\tan^{-1}\left(\frac{3y}{x}\right)=x+c$

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$$\tan^{-1}\left(\frac{3y}{x}\right)=3x+c$$

D) $$\frac{-\phi(x)dy+y\phi'(x)dx}{y^2}=dx$$

$$\int d\left(\frac{\phi(x)}{y}\right)=\int dx$$

$$\frac{\phi(x)}{y}=x+c$$

$$\phi(x)=y(x+c)$$

43. $n(S)>1$ implies that M is singular

$$N+3I=MN+2M$$

$$\Rightarrow MN+2M-N-2I.....(1)$$

$$\Rightarrow (M-I)(N+2I)=I\Rightarrow M-I \text{ and } N+2I \text{ are invertible}$$

$$\Rightarrow (N+2I)(M-I)=I\Rightarrow NM-N+2M-2I=I.....(2)$$

$$MN=NM\Rightarrow MN^2=N^2M$$

44. $$\mathrm{N}=\sum_{r=0}^{100}(-1)^r.4^{100-r}.{}^{200}\mathrm{C}_r.{}^{200-r}\mathrm{C}_{100-r}$$

$$=\sum_{r=0}^{100}(-1)^r.4^{100-r}.\frac{200!}{r!(200-r)!}\frac{(200-r)!}{100!}$$

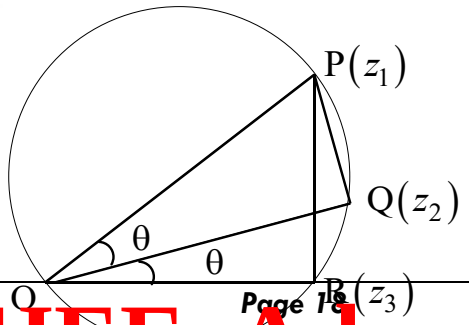
$$={}^{200}\mathrm{C}_{100}\sum_{r=0}^{100}(-1)^r\,4^{100-r}.{}^{100}\mathrm{C}_r$$

$$=\mathrm{N}=3^{100}.{}^{200}\mathrm{C}_{100}$$

45. $$2\sqrt{3}z_2^2=\left(2+\sqrt{3}\right)z_1z_3$$

$$2\sqrt{3}=\left(2+\sqrt{3}\right)\frac{1}{\cos\theta}\times\frac{\cos 2\theta}{\cos\theta}$$

$$2\sqrt{3}\left(2-\sqrt{3}\right)=\frac{2c^2-1}{c^2}=\left(2-\frac{1}{c^2}\right)$$



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$$\frac{1}{c^2}=2-2\left(2\sqrt{3}-3\right)=2\left(4-2\sqrt{3}\right)$$

$$\frac{1}{c^2}=4\left(2-\sqrt{3}\right)$$

$$\cos^2=\frac{2+\sqrt{3}}{4}=\frac{\left(\sqrt{3}+1\right)^2}{8}$$

$$\cos\theta=\frac{\sqrt{3}+1}{2\sqrt{2}}$$

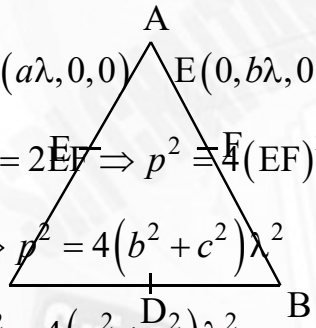
$$=\cos\left(\frac{\pi}{4}-\frac{\pi}{6}\right)$$

$$=\cos\left(\frac{\pi}{12}\right)$$

$$\alpha^0=15$$

$$\frac{\alpha}{3}=5$$

46. Key : (2.00)



$$D(a\lambda,0,0)\quad E(0,b\lambda,0)\quad F(0,0,c\lambda)$$

$$P=2EF\Rightarrow p^2=4(EF)^2$$

$$\Rightarrow p^2=4\left(b^2+c^2\right)\lambda^2$$

$$q^2=4\left(a^2+c^2\right)\lambda^2$$

$$r^2=4\left(a^2+b^2\right)\lambda^2$$

$$p^2+q^2+r^2=8\lambda^2\left(a^2+b^2+c^2\right)$$

$$p^2+q^2+r^2=8\lambda^2\left(\frac{a^2+b^2+r^2}{2}\right)\Rightarrow\lambda^2=1/4$$

47. Equation of tangent at $(x_i,\ y_i)$ on the curve $y=f(x)$ is $Y-y_i=m(X-x_i)$

Put $Y=0,\ X=x_{i+1}$ then $0-y_i=m\left(x_{i+1}-x\right)\Rightarrow x_{i+1}-x_i=-\frac{y_i}{m}$

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$\therefore \frac{x_{i+1}}{x_i} - 1 = -\frac{y_i}{mx_i}$

If $x_1, x_2, x_3, \ldots\ldots\ldots x_n$ form G.P. then $r - 1 = -\frac{y_i}{mx_i}$ (where r is common ration of G.P)

$\therefore m = \frac{dy}{dx} = -\frac{y}{(r-1)x} \Rightarrow \frac{dy}{y} + \frac{dx}{(r-1)} = 0$

$\Rightarrow \ell ny + \left(\frac{1}{r-1}\right)\ell nx = k$

$\Rightarrow y(x)^{\frac{1}{r-1}} = c$

$\Rightarrow r = 2,\ xy = c,\ e = \sqrt{2}$

48. $4I + 3J = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$

$3I - 4J = \int_0^{\pi/2} \frac{3\cos x - 4\sin x}{3\sin x + 4\cos x + 25} dx$

$= \left[\ln(3\sin x + 4\cos x + 25)\right]_0^{\pi/2} = \ln 28 - \ln 29$

$3I - 4J = \ln\left(\frac{28}{29}\right) \ldots\ldots\ldots (i)$

$4I + 3J = \frac{\pi}{2} \ldots\ldots\ldots\ldots\ldots (ii)$

$\Rightarrow$ d = 1 and b = 1, c = 28, d = 29

$\therefore a + b + c + d - 55 = 59 - 55 = 4$

49. $(x+1)^3 + (y+1)^3 = 16$

$y = -1 + \left(16 - (x+1)^3\right)^{1/3} = f(x)$

$x = -1 + \left(16 - (y+1)^3\right)^{1/3}$

$\therefore f^{-1}(x) = -1 + \left(16 - (x+)^3\right)^{1/3} = g(x)$

$\Rightarrow f(x) = g(x)$

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$f\left(-1+\sqrt[3]{15}\right)=g\left(-1+\sqrt[3]{15}\right)=0$

$f\left(x+g(x).g\left(x+f(x)\right)\right)=\left(f\left(x+f(x)\right)\right)^2$

$\frac{d}{dx}f\left(f\left(x+f(x)\right)\right)^2=2f\left(x+f(x)\right)$

$f'\left(x+f(x)\right).\left(1+f(x)\right)$

$\frac{d}{dx}\left(f\left(x+f(x)\right)\right)^2$ at $x=-1+\sqrt[3]{15}=0$

$\because g\left(-1+\sqrt[3]{15}\right)=f\left(-1+\sqrt[3]{15}\right)=0$

50. $a\left(x^3-1\right)+(x-1)=0$

$(x-1)\left(ax^2+ax+a+1\right)=0$

$\alpha, \beta \neq 1$

$ax^2+ax+a+1=0$

$\alpha+\beta=-1,\ \alpha\beta=\frac{a+1}{a}$

$$\lim_{x\to\frac{1}{a}}\frac{(1+a)x^3-x^2-a}{\left(e^{1-ax}-1\right)(x-1)}=\lim_{x\to\frac{1}{a}}\frac{\left(x^3-x^2\right)+a\left(x^3-1\right)}{\left(e^{1-ax}-1\right)(x-1)}$$

$$\lim_{x\to\frac{1}{a}}\frac{\left[x^2+a\left(x^2+x+1\right)\right]}{\left(e^{1-ax}-1\right)}=\lim_{x\to\frac{1}{a}}\frac{(1+a)x^2+ax+a}{\left(\frac{e^{1-ax}-1}{1-ax}\right)(1-ax)}$$

$$=\lim_{x\to\frac{1}{a}}\frac{\left[\left(\frac{1+a}{a}\right)\right]}{(1-ax)}=\lim_{x\to\frac{1}{a}}\frac{\left(\alpha\beta x^2-(\alpha+\beta)x+1\right)}{(1-ax)}$$

$$=\lim_{x\to\frac{1}{a}}\frac{(1-(\alpha)x)(1-(\beta)x)}{(1-ax)}=\frac{a(\alpha-\beta)}{\alpha}$$

51. Length of $PQ=TCT=\sqrt{9^2-(6-3)^2}=6\sqrt{2}$

Let $C_1: x^2+y^2=9^2$

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Let $\angle QTN = \theta$

$\therefore \frac{NT}{MT} = \frac{r_3}{r_2} = \frac{3}{6} = \frac{1}{2} = \frac{TQ}{TP} \Rightarrow N$ is mid point of TM

$\Rightarrow TP = 2TQ \Rightarrow TQ = PQ$

$\Rightarrow TP = 6\sqrt{2}$

$\therefore NT = \sqrt{72+9} = 9$

$\Rightarrow LT = 9 + (9-3) = 15$

$\because L(0,0) \Rightarrow T(-15,0)$

$\sin\theta = \frac{NQ}{NT} = \frac{3}{9} = \frac{1}{3}$

$\Rightarrow \cos\theta = \frac{6\sqrt{2}}{9} = \frac{2\sqrt{2}}{3}$

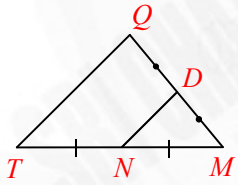
Any point on chord AB be $\left(-15 + \frac{2\sqrt{2}r}{3}, \frac{r}{3}\right)$

Lie on circle $C_1 : x^2 + y^2 = 81$

$\Rightarrow r^2 - 20\sqrt{2}r + 144 = 0$ (r_1, r_2)

$\therefore AB = |r_1 - r_2| = \sqrt{(r_1+r_2)^2 - 4r_1r_2}$

$= \sqrt{800-576} = \sqrt{224} \qquad = 4\sqrt{14}$



$\because N$ is mid-point of Tm and D is mid-point of QM

$\Rightarrow ND = \frac{1}{2}TQ \qquad = \frac{1}{2}\left(6\sqrt{2}\right) = 3\sqrt{2}$

52. Length of $PQ = TCT = \sqrt{9^2 - (6-3)^2} = 6\sqrt{2}$

Let $C_1 : x^2 + y^2 = 9^2$

Let $\angle QTN = \theta$

$\therefore \frac{NT}{MT} = \frac{r_3}{r_2} = \frac{3}{6} = \frac{1}{2} = \frac{TQ}{TP} \Rightarrow N$ is mid point of TM

$\Rightarrow TP = 2TQ \Rightarrow TQ = PQ$

$\Rightarrow TQ = 6\sqrt{2}$

$\therefore NT = \sqrt{72+9} = 9$

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53,54. $f(x)=\begin{cases}-2; & -1\le x<-\frac{1}{2}\\ -1; & -\frac{1}{2}\le x<0\\ 0; & 0\le x<\frac{1}{2}\\ 1; & \frac{1}{2}\le x<1\\ 2x; & 1\le x\le 2\end{cases}$

$f(g(x))=\begin{cases}-2; & x\in\left[\frac{7\pi}{6},\frac{11\pi}{6}\right]-\left\{\frac{3\pi}{2}\right\}\\ -1; & x=\frac{3\pi}{2}\\ 1; & x\in\left(\frac{\pi}{6},\frac{5\pi}{6}\right)\\ \operatorname{cosec} x; & x\in\left(0,\frac{\pi}{6}\right]\cup\left[\frac{5\pi}{6},\pi\right)\end{cases}$

$f(g(x))=\begin{cases}-2; & -\frac{1}{2}\le x<0\\ 0; & 0\le x<\frac{1}{2}\\ 2; & \frac{1}{2}\le x<1\\ 4; & x=1\end{cases}$

$g(f(x))=\begin{cases}\frac{\operatorname{cosec}1}{2}; & \frac{1}{2}\le x<1\\ \frac{\operatorname{cosec}2x}{2}; & 1\le x\le 2-\left\{\frac{\pi}{2}\right\}\end{cases}$

$f(2g(x))=\begin{cases}-1; & x=\frac{3\pi}{2}\\ 2\operatorname{cosec} x; & x\in\left[\frac{\pi}{6},\frac{5\pi}{6}\right]\end{cases}$

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